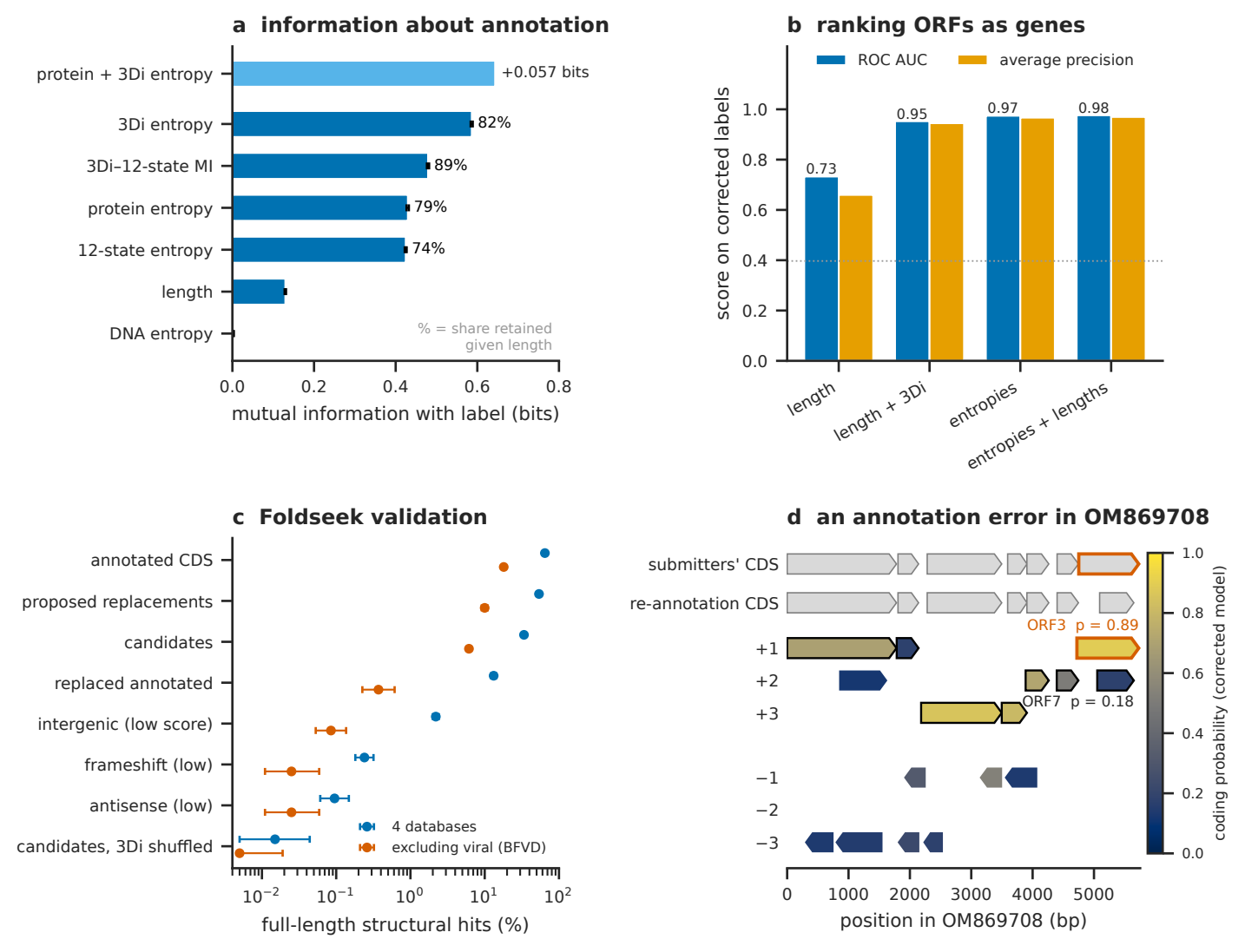
a information about annotation
protein + 3Di entropy
+0.057 bits
3Di entropy
82%
3Di–12-state MI
89%
protein entropy
79%
12-state entropy
74%
length
DNA entropy
% = share retained given length
0.0
0.2
0.4
0.6
0.8
mutual information with label (bits)
b ranking ORFs as genes
ROC AUC
average precision
score on corrected labels
0.0
0.2
0.4
0.6
0.8
1.0
0.73
0.95
0.97
0.98
length
length + 3Di
entropies
entropies + lengths
c Foldseek validation
annotated CDS
proposed replacements
candidates
replaced annotated
intergenic (low score)
frameshift (low)
antisense (low)
candidates, 3Di shuffled
4 databases
excluding viral (BFVD)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
full-length structural hits (%)
d an annotation error in OM869708
submitters' CDS
re-annotation CDS
ORF3 p = 0.89
+1
+2
+3
ORF7 p = 0.18
−1
−2
−3
0
1000
2000
3000
4000
5000
position in OM869708 (bp)
coding probability (corrected model)
0.0
0.2
0.4
0.6
0.8
1.0